
Stock Portfolio Modelling: Time Series and Monte Carlo Methods for Portfolio Optimisation

K. Kapoor*

University of Southampton, Southampton, United Kingdom

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Financial markets are inherently volatile and unpredictable. We can utilise the stochastic behaviour of markets to forecast stock prices and returns, from which we can generate an optimal portfolio that ensures the optimal expected return from the portfolio relative to the risk of investment. This project utilises time series analysis of past stock returns to forecast future returns. ARIMA models are used to explore the behaviour of mean returns, while GARCH models are used to forecast the volatility of returns. Following this, Monte Carlo simulations are used from the extracted parameters of the GARCH model to generate possible paths that the stock price can follow. We then use the framework of Modern Portfolio Theory to generate optimal portfolio weights for a given portfolio of 5 stocks by maximising the Sharpe Ratio. This study aims to evaluate the effectiveness of risk-adjusted portfolio allocation while exploring the practical limitations enforced by non-stationary markets.

I. INTRODUCTION

A. Purpose

This project aims to investigate financial modelling techniques to construct an optimised investment portfolio given a group of assets (stocks). Financial markets are often governed by complex interactions alongside stochastic processes, which makes accurate financial prediction rather difficult, as markets are also prone to shocks and non-stationary trends. This study combines forecasting models with volatility analysis and simulation techniques to analyse past market data and predict possible future market trends via random walks and Monte Carlo methods. We then use numerical methods to maximise the Sharpe ratio to obtain an optimal portfolio.

B. Background

Financial markets are influenced by a multitude of factors, from geopolitical events to investor sentiments to company performance, which results in a complex dynamic system. As a consequence of this, stock prices and returns from investments fluctuate constantly. Traditional investment strategies that focus on intuition or maintaining a diverse portfolio (that assumes independent asset movement) may not always account for the risk of investment in more volatile periods of the market.

Modern Portfolio Theory introduced a mathematical framework that allows for balancing the risk and expected return, and

utilises covariance between assets to account for relationships and movement between assets as well.

ARIMA models are used to analyse trends and autocorrelation between time series for mean returns, but require a constant variance. Meanwhile, ARCH models such as GARCH are used to capture volatility trends in markets due to their assumption of time-dependent variance, ensuring that random shocks to a system can be modelled effectively.

Random walks can be initialised from the last known state, where shocks are applied to the system to represent possible volatile market trends. These random walks are simulated many times over a forecasting period to account for possible ways the price of the stock will evolve.

These methods allow us to analyse market behaviour and construct a portfolio where each asset's weightage allows us to maximise return while controlling the investment risk.

C. Objectives

The primary objectives of this project are as follows:

- Collect and process stock market data for a portfolio of selected stocks.
- Analyse behaviour of returns and use time series analysis methods to justify model choices for forecasting.
- Implement ARIMA models to forecast mean stock returns and GARCH models to forecast return volatility.
- Use Monte Carlo Simulations in tandem with mean returns and random shocks modelled from the volatility of stock returns.
- Construct and optimise an investment portfolio of the selected stocks and evaluate the risk-adjusted measures of performance.

* Contact author: krishk122703@gmail.com

- Investigate practical limitations to predictive modelling in dynamic financial markets.

D. Scope

This project focuses on the analysis and optimisation of a portfolio consisting of 5 publicly traded stocks using historical daily market data obtained from the Yahoo Finance Public API. This study is also limited to statistical and stochastic modelling approaches such as Geometric Brownian Motion, ARIMA forecasting and GARCH modelling alongside Random Walks and Monte Carlo simulation.

The project does not aim to predict the exact future of financial markets with certainty, nor does it account for any transaction costs or any macroeconomic or foreign trade policies. This project also only focuses on buying stocks expecting them to rise in the long term (“going long”) and does not entertain ideas of short selling (selling stock at a price to buy back at a lower price). As such, the portfolio optimisation methods **should not be viewed as financial advice**, as this project has been developed for academic purposes only.

II. THEORY

A. Modern Portfolio Theory

$$R = \frac{V_f - V_i}{V_i} \quad (1)$$

Modern Portfolio Theory is a theory by which investment portfolios have to maximise portfolio return while trying to minimise portfolio risk. This theory relies on the idea that a diverse portfolio that takes into consideration the movement of portfolio assets together, alongside market behaviours, can help reduce the total risk of a portfolio without significantly impacting the returns. Relations between assets are considered when tracing the movement of the portfolio through mathematical tools such as the covariance matrix and correlation matrix, which help quantify relations between assets. As such, modern portfolios tend to have a diverse array of investments that are generally not focused on one sector, to allow for less risk.

One of the central ideas of the Modern Portfolio Theory is the idea of a risk-return tradeoff; for a high return portfolio, investors must accept a higher risk of investment, while a lower risk of investment results in a lower return portfolio. The portfolio risk is often adjusted from the volatility (standard deviation) of the portfolio returns. By adjusting the weightings of the assets in the portfolio, investors can achieve a balance between the volatility of the stock and its expected return, which results in an optimal portfolio. [1]

The expected return of a portfolio is given by:

$$E(R_p) = \sum_{i=1}^n w_i E(R_i) \quad (2)$$

Where $E(R_p)$ is the portfolio expected return, w_i is the weightage of the individual asset and $E(R_i)$ is the expected return of the individual asset. The variance of the portfolio can be shown as follows:

$$\sigma_p^2 = w^T \Sigma w \quad (3)$$

Where σ_p^2 is the variance of the portfolio, w is a vector containing the portfolio weights, w^T is the transpose of the weight vector, and Σ is the covariance matrix between assets. The risk can be taken as the square root of the variance σ_p , which represents the volatility of the portfolio.

One of the key consequences of the Modern Portfolio Theory is the efficient frontier, which is often a set of portfolios that represent the highest possible return at the given amount of risk. These are often considered the most optimal portfolios for the level of risk.

The performance of portfolios is often represented by the Sharpe Ratio, which is given as follows:

$$S_p = \frac{E(R_p) - R_f}{\sigma_p} \quad (4)$$

Where S_p is the Sharpe ratio, $E(R_p)$ is the expected return of the portfolio, R_f is the risk-free return rate (which for this paper is 0.0), and σ_p is the portfolio volatility.

B. Time series forecasting

Time series forecasting is often used when trying to predict future financial data, as historical financial data is used to draw upon future predictions through forecasting methods. However, financial time series data is inherently noisy and exhibits stochastic (probabilistic) behaviour alongside constantly changing trends. As a consequence of this, predictions become increasingly difficult.

We can utilise the concept of autocorrelation to help determine future data. Autocorrelation is a method by which past values may influence future observations. We can often use autocorrelation function plots (ACF) or partial autocorrelation function plots (PACF) to determine if autocorrelation is present within the data; such plots represent the relation between the plot and a lagged version of the plot and are used to indicate if previous values do affect present or future values.

Oftentimes, instead of modelling stock prices directly, it is more common to use logarithmic returns (often referred to in this paper as log returns) of the stock price. Log returns tend to be more statistically stable and maintain properties such as being additive across time intervals, which makes them easier to calculate over long time frames. One can calculate the log returns of a stock by the following equation:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \quad (5)$$

Where r_t is the logarithmic return at time t , P_t is the price at time t , and P_{t-1} is the price at time $t-1$.

The AutoRegressive Integrated Moving Average (ARIMA) model is used for financial forecasting of mean returns due to its ability to model moving average behaviour with time series data. A general ARIMA model can be expressed by the following equation:

$$y'_t = c + \sum_{i=1}^p \varphi_i y'_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j} \quad (6)$$

where p represents the autoregressive term (it defines the number of lag terms used to predict future values), q represents the moving average term (this defines the size of the moving average window), while ε_t represents the residual error term (representing the difference between predicted and observed values).

The ARIMA model relies on 3 major assumptions for it to be useful in predictive capability. The mean should be constant, the variance of residuals should be constant (homoskedasticity), and there should be autocorrelation present. For the condition of autocorrelation, the usage of ACF plots is necessary, as they show if autocorrelation is present. [2]

C. Volatility Modelling

Financial returns often display periods of high volatility and, conversely, periods of low volatility that are clustered together; this is known as volatility clustering. This behaviour is in opposition to the requirement for ARIMA models, which is a constant variance (homoskedasticity). As such, financial markets often display a phenomenon where variance is heavily time-dependent (heteroskedasticity).

Volatility modelling is important as financial risk is directly related to the volatility of the market; as such, it needs to be considered heavily in case of events where markets become erratic or asset pricing differs due to external factors.

The Generalised AutoRegressive Conditional Heteroskedasticity (GARCH) model is utilised for financial markets, where modelling a time-dependent variance is necessary. GARCH(1,1) is a popular model for forecasting volatility, which utilises previous volatility and previous residual shocks to estimate the current volatility. In the case of a constant mean, we can utilise the log returns instead of extracting the ARIMA residuals. It can be represented by the following equation:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (7)$$

Where σ_t^2 represents the conditional variance ε_{t-1}^2 represents the squared residual shock from the previous period. While α , β , and ω are model parameters. GARCH models also rely on the knowledge of the distribution of the input to understand the behaviour of volatility across time, oftentimes Normal Distributions or the Student's T distribution (as it accounts

for wider tails, allowing for extreme events to be shown more commonly) are used to account for the distribution of returns. [3]

GARCH models can then be used to generate random shocks to account for such extremes when simulating stock prices.

D. Monte Carlo Methods

Monte Carlo Simulation is a computational method by which we can use repeated random sampling to model uncertain systems and estimate possible outcomes. Monte Carlo methods are often used to simulate stock prices when markets display stochastic behaviour.

Monte Carlo methods allow us to produce several different forecasts of future price paths based on expected mean returns and forecasted volatility produced by the ARIMA and GARCH models.

We can then use the volatility obtained by the GARCH models alongside distributions such as the Normal and Student's T distribution. From the GARCH model's forecasted volatility and the last known price, we can extract the log returns from the following formula:

$$r_t = \mu + \sigma_t z_t \quad (8)$$

Where r_t is the log returns of the data, μ is the forecasted mean return of the data, σ_t is the volatility/standard deviation forecasted by the GARCH model, and z_t is the distribution by which we are modelling the returns (or residuals if your mean varies).

Once log returns are obtained, they can be converted to stock prices over the forecasted time period through the following equation:

$$P_t = P_0 \exp\left(\sum_{i=1}^t r_i\right) \quad (9)$$

Where P_t is the stock price, P_0 is the last known stock price before forecasting, and the exponential term represents the sum of the logarithmic terms.

By combining Monte Carlo methods with volatility forecasts generated from the GARCH models, we can observe the effect of generating distributions that have random market changes accounted for, as well as allowing for a more accurate mathematical approximation to the evolving market.

We can use Geometric Brownian Motion (GBM) as a comparative stochastic process, which relies on a constant mean, constant volatility, stochastic shocks and a deterministic drift, as a benchmark method to generate similar paths that can be compared to the GARCH forecasts. The formula for Geometric Brownian Motion is shown below:

$$P_t = P_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma \sum_{i=1}^t \varepsilon_i \sqrt{\Delta t}\right) \quad (10)$$

Where P_t represents the price at time t , P_0 represents the last known price, μ and σ represent the respective expected mean and volatility/standard deviation (the random stochastic shock), accompanied by the summation of $\sum_{i=1}^t \varepsilon_i \sqrt{\Delta t}$, which represents the Wiener process that is used to model white noise, noisy data that has a constant mean and variance but no autocorrelation.

III. DATA AND PREPROCESSING

A. Data Collection

Market data was collected for a portfolio of consisting of 5 publicly traded stocks. The selected stocks used in the study (along with relevant ticker symbols) are as follows:

- Advanced Micro Devices (AMD)
- Alphabet Inc. (GOOGL)
- JPMorgan Chase & Co. (JPM)
- Exxon Mobil Corporation (XOM)
- The Coca-Cola Company (KO)

This portfolio spans across technology, finance, consumer goods and the energy sector, allowing for a diverse portfolio of stocks.

Daily historical data has been used from 02nd January 2020 to 09th May 2026, obtained from Yahoo Finance's Python public API. The opening price and adjusted closing price data are loaded onwards.

Financial datasets commonly contain irregularities such as trading holidays, days where entries were not recorded, and any accidental negative value days. Despite Yahoo Finance's ability to avoid this, it is always best to discuss and clean the data for any such irregularities. As well as this, any major financial irregularities, such as after-hours trading, are to be investigated, as well as any days where price fluctuations heavily exceed 3 standard deviations of the trading month's (21 trading days) rolling mean.

Once all irregularities are removed and any missing data is filled through forward fill methods, the opening price data is dropped, and only the adjusted closing price data is retained for this project. An example of the first 5 rows of the dataframe is shown below:

Date	AMD	GOOGL	JPM	KO	XOM
2020-01-02	49.10	67.87	118.43	45.43	53.31
2020-01-03	48.60	67.52	116.87	45.18	52.88
2020-01-06	48.39	69.32	116.77	45.17	53.28
2020-01-07	48.25	69.18	114.79	44.82	52.85
2020-01-08	47.83	69.68	115.68	44.90	52.05

TABLE I. Stock closing prices (USD) for selected companies (rounded prices), January 2020

The cleaned stock price data is then plotted over time, which is shown below:

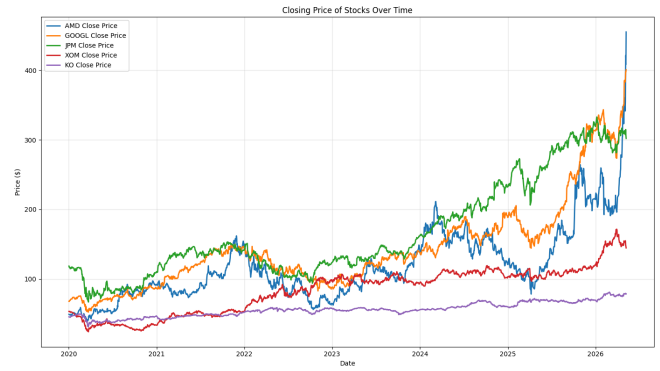


FIG. 1: Cleaned Stock Data Price (\$) to Date for all stocks in the test portfolio

B. Returns

Direct stock prices are unsuitable for statistical forecasting due to their non-stationary data that exhibits strong trends which change drastically over time. Log return data is then generated using Eq. (5) and added to the dataframe.

Log returns offer far more advantage than raw prices or simple percentage returns. Log returns help stabilise variance and reduce long-term exponential growth. As well as this, log returns often follow a normalised distribution, which makes them extremely compatible with financial modelling techniques, including ARIMA and GARCH models.

We can visualise historical log return plots against time (alongside rolling mean and rolling standard deviation), as well as a histogram showing the frequency of log returns and a normal distribution overlaid over the log returns. The plots for which are shown below for AMD's stock data:

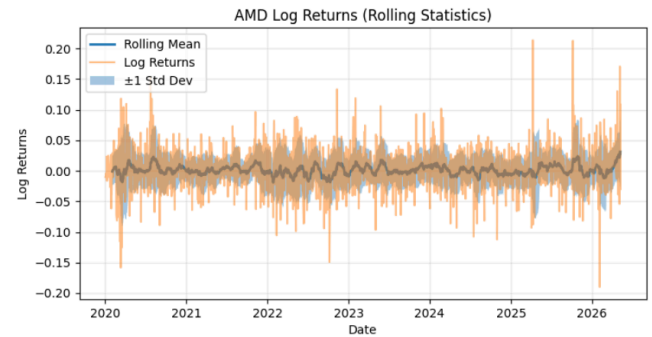


FIG. 2: Log Returns vs Date for AMD with rolling 21-day mean and standard deviation

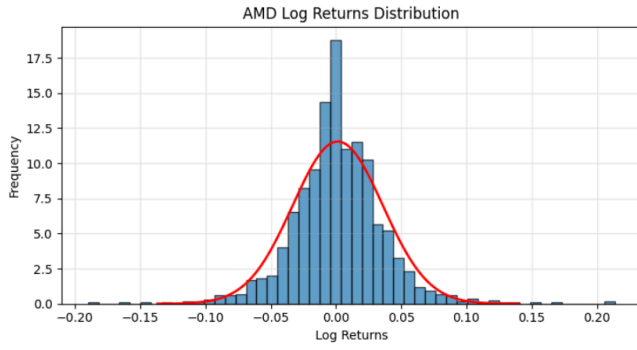


FIG. 3: Histogram of Frequency vs Log Returns for AMD overlaid with a plot of the normal distribution

C. Statistical Analysis

Following the preprocessing and return calculations, statistical analysis techniques are performed to better understand the behaviour of the financial data.

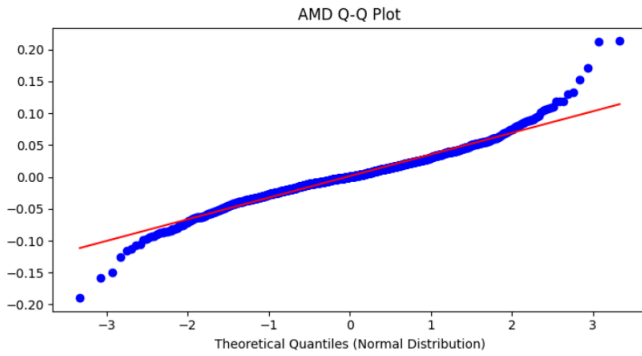


FIG. 4: Quantile-Quantile (QQ) plot of log returns against the normal distribution, used to show behaviour at tail ends

A quantile-quantile (QQ) plot is used to show the behaviour of the log returns data in comparison to the normal distribution. As shown by the QQ plot for AMD's log returns above, we see that the tail end behaviour for the model is evidently different from the straight line presented by the normal distribution. This vastly different behaviour of the tail ends leads to the motivation for using wider tail distributions, such as the Student's t distribution, which has an increased kurtosis (wider tails).

Another important statistical analysis that needs to be discussed is the correlation matrix. In Modern Portfolio Theory, closely correlated stocks tend to move similarly (due to the dependence on the covariance matrix), as such a correlation matrix is necessary to see the correlation between certain stocks, as shown below:

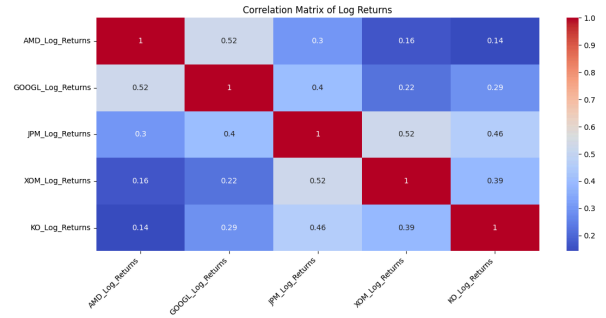


FIG. 5: Correlation Matrix showing the relation between log returns of plots relative to another (indicating level of correlation)

We see that AMD and Google hold a correlation coefficient of 0.53, which indicates a close correlation, resulting in the idea that these 2 stocks will move similarly and the reasoning for their correlation is imminent, as both companies invest heavily in hardware development and artificial intelligence products.

IV. METHODOLOGY

A. Forecasting Methods

The ARIMA model is applied to the log returns of each stock in the training data. The data is then split into training and test data; the training data is for training the models used for training the ARIMA and GARCH models. This approach allows for model performance to be assessed on unseen data.

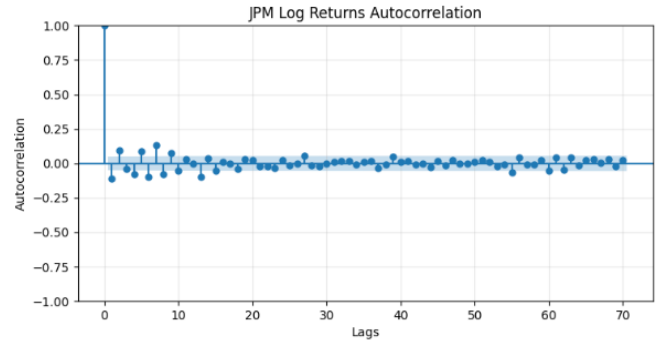


FIG. 6: Autocorrelation vs Time lags plot for JPMorgan log returns (ACF plot).

As seen by the ACF plot of log returns against the lags for JPMorgan Chase & Co., the time series data for the log returns does not show any autocorrelation, as most of the data is centred around 0 for autocorrelation (with an outlier near 2020 when markets were heavily impacted by COVID-19). This implies that an ARIMA model is likely to encounter white noise. The ARIMA model used is selected via Python's `pmdarima` module through the `auto_arima` function, which results in selecting `ARIMA(1,0,0)`.

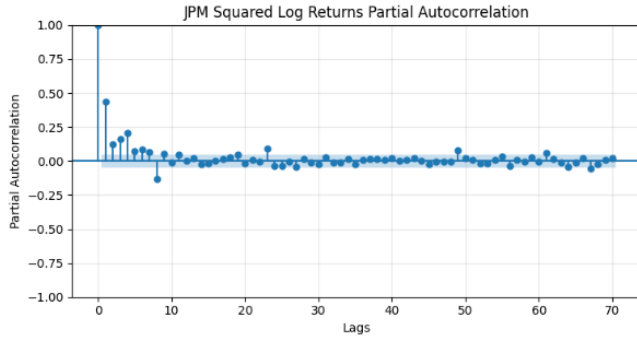


FIG. 7: Partial Autocorrelation vs Time lags plot for JPMorgan squared Log returns (ACF plot).

The PACF plot of squared log returns shows a decaying relation over time, implying a time-varying variance. Amidst this and the plot shown by Fig. 2, we see clearly that the volatility heavily varies with time, while the mean does not as much. As such, the data itself is heteroskedastic (time varying dependence), making a solely ARIMA-centred model not the ideal approach, as volatility needs to be modelled and forecasted. GARCH models need to be utilised to forecast volatility, while the mean is forecasted by the ARIMA model.

The mean is forecasted by the ARIMA(1,0,0) model on the training data and then forecasts the mean returns 30 days ahead.[2]

B. Volatility Forecasting

Volatility forecasting is performed using GARCH(1,1), which is fitted on the log returns data. It is used for its effectiveness in capturing general trends in volatility from which we can obtain the conditional variance, as shown by Eq. (7) and the requested modelling distribution. A student's t distribution is used to model the log returns to allow for emphasis on an increased kurtosis dataset (the option for the use of the normal distribution is available within the config.py file of the repository).

Following the model fitting, the model is incrementally forecasted from the last known to forecast the next 30 days (forecast horizon). It is then plotted against the actual log returns while model parameters such as α , β and γ are extracted and saved as pickle files to later assist in simulations alongside the forecasted volatility. [3]

C. Monte Carlo Simulation

Monte Carlo simulations are run using the extracted mean from the ARIMA model as well as the volatility generated by the forecasted GARCH data.

A random walk is generated by the forecasted volatility of the data, the utilised distribution (from which we modelled the log returns data of the GARCH model), as used in Eq. (8).

A set of specified simulations (10000) is run, of which 50 are then plotted to generate simulated paths of the price data from which we can also generate a confidence interval between 5%

and 95% to represent where the vast majority of the forecasted price is distributed.

D. Portfolio Optimisation Methods

Portfolio optimisation is used by calculating the expected return of the portfolio (Eq. (2)) and portfolio risk (Eq. (3)), which requires the usage of the covariance matrix to dictate how markets can allow the movement of correlated stocks.

The annual (assuming 252 trading days in a year) return for the portfolio can be calculated by the following equation:

$$E(R_{\text{annual}}) = (1 + E(R_{\text{daily}}))^{252} - 1 \quad (11)$$

The annualised volatility of the portfolio (annualised portfolio risk) can be calculated as follows:

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} * \sqrt{252} \quad (12)$$

We can optimise the portfolio by trying to maximise the Sharpe Ratio (assuming a 0.0 risk-free return rate), which we can calculate using Eq. (4). Similar to the annualised volatility, we can annualise the Sharpe ratio by the following equation:

$$S_{\text{annual}} = S_{\text{daily}} * \sqrt{252} \quad (13)$$

To allow for a maximised Sharpe ratio, we can reframe our condition to minimise the negative Sharpe ratio, subject to the following constraints and bounds:

$$\sum_{i=1}^n w_i = 1 \quad (14)$$

Where w_i is the weightage of each asset. This equation states that all the weights added should sum to 100% of the capital you allocate between n assets.

$$0 \leq w_i \leq 1 \quad (15)$$

Provided no short-selling is allowed, this bound states that each asset weight must be between 0 and 1 to not exceed the capital added into the portfolio.

We can then use numerical minimisation solving methods such provided by scipy. optimise, such as Sequential Least Squares Programming methods, which rely on separating complex systems into subsystems which are solvable by Least Squares Programming methods. This results in a weightage allocation where the Sharpe ratio is at its highest, implying the largest allowed value for portfolio return relative to the portfolio risk.[4]

V. RESULTS

A. Forecasting Results

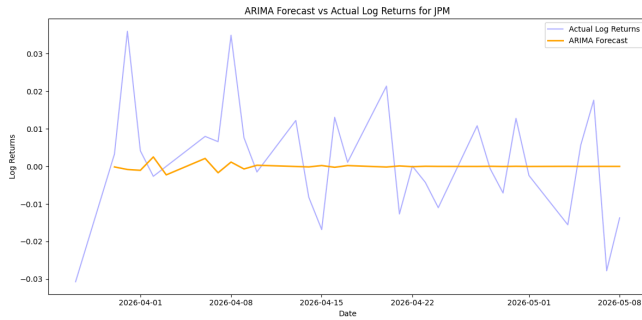


FIG. 8: Mean log returns forecasted for next 30 days by ARIMA(1,0,0) model

ARIMA forecasting models were fitted to the log return data for each stock symbol from the training set individually, and each was evaluated on the test set. We can see the results of the mean forecasting for 30 days above. We see from this plot that our ARIMA model predicts the mean to be constant (approximately 0) and stationary across the 30 forecasted days. Meanwhile, the actual log returns for the 30 days of the training set display a large volatility; even if the mean is constant, the model cannot predict the volatile nature of the log returns, and that autocorrelation is not present within this structure (as also shown by Fig. 6). The mean forecasted log return of 0 can be extracted for its use in the GARCH model and Monte Carlo simulations.

B. Volatility Results

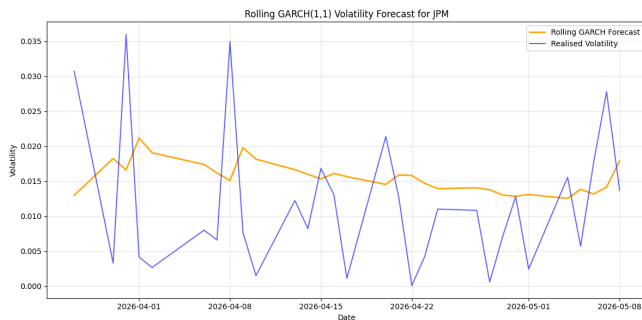


FIG. 9: Forecasted Volatility against Date for GARCH(1,1) model

GARCH volatility forecasting models can be used to forecast the results, as we have seen the ARIMA model provide a constant mean of zero for log returns. Once GARCH is utilised to forecast the volatility, while we see a model whose mean still centres around 0, we see that the GARCH model captures the volatile nature of the log returns, especially the overall structure of the data, even if the magnitude of the log returns is not as correct. This helps ensure a more robust model structure that can be used to model shocks to the system effectively.

C. Monte Carlo Results

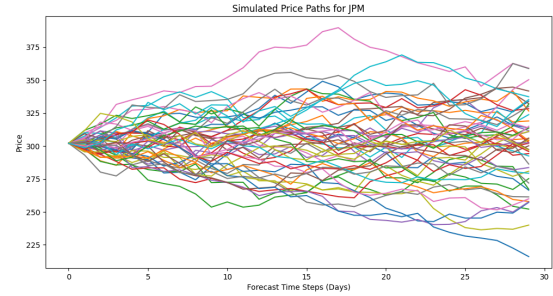


FIG. 10: Monte Carlo Simulations of price paths across the next 30 days through geometric brownian motion. Key Assumptions: Constant Mean and constant volatility

Geometric Brownian Motion, assuming a constant volatility and constant mean, allows for a statistical forecasting of results remains relatively stable and the price of JPM shares are confined within the price range of \$275 and \$350 in most cases with some extremes between \$225 and \$375; however, due to this, the model isn't able to react towards extreme shocks within the system, meaning extreme trends such as high price rises or drops are not as easily predicted by the Geometric Brownian Motion model. This makes for a very reasonable estimate when no major factors affect the market or individual stocks.

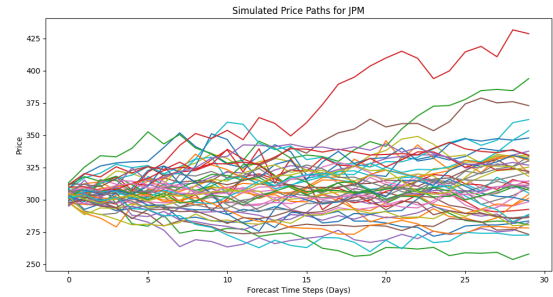


FIG. 11: Monte Carlo Simulations of price paths across the next 30 days. Key assumptions: assuming a normal distribution of returns, constant mean and varying volatility.

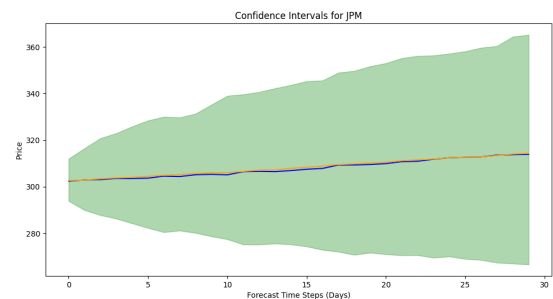


FIG. 12: Confidence intervals (between 5 and 95% of paths) for Monte Carlo Simulations of price paths across the next 30 days. Key assumptions: assuming a normal distribution of returns, constant mean and varying volatility.

The figures above show the Monte Carlo simulations for stocks assuming the log returns are normally distributed, as well as a plot of the confidence interval within which 90% of the data lies as well as the most stable path that ends with a share price of \$310. This once again shows that most stock prices for JPM are relatively stable in the range between \$270 and \$380; however, the GARCH model allows for more extreme shocks ranging between \$200 and \$425 in the market, which seem relatively unrealistic but, in cases of volatile markets, are completely plausible, and these extremes are considered when optimising for our portfolio.

D. Portfolio Optimisation Results

The efficient frontier is a set of optimal portfolios which tend to obtain the highest level of expected return based on the relative investment risk (which should be minimised). The plot below shows a portfolio risk vs return plot for a set of generated portfolios, where the most efficient portfolios tend to be lighter in colour (implying a higher Sharpe ratio).

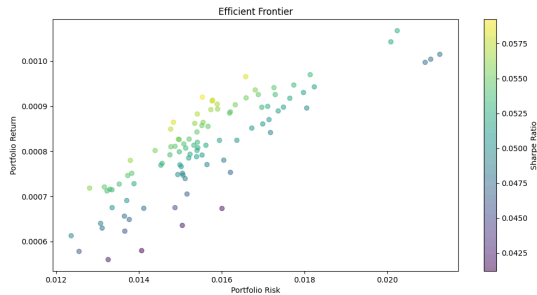


FIG. 13: Efficient Frontier plot for portfolios. Portfolio Risk vs Expected Portfolio Return.

This plot implies a daily Sharpe ratio of around 0.0580 is the highest Sharpe Ratio possible, which forms the most optimal portfolio at a daily portfolio risk of 0.00093 with a return of around 0.0160.

Metric	Value
AMD Weight	0.1186
GOOGL Weight	0.5340
JPM Weight	0.0000
XOM Weight	0.2124
KO Weight	0.1349
Portfolio Return	0.0009
Annualised Return	0.25445
Portfolio Risk	0.0158
Annuaised Risk	0.25082
Daily Sharpe Ratio	0.0595
Annualised Sharpe Ratio	0.9439

TABLE II. Optimal Portfolio Allocation as well as Portfolio Risk, Portfolio Return and Sharpe ratio metrics

The table above shows the portfolio weights for the most optimal portfolio (i.e., the highest daily Sharpe ratio) where JPMorgan Chase & Co. seems to have no asset allocation, implying it bears the least influence on maximising the Sharpe ratio. Meanwhile, a relatively stable stock like Alphabet Inc. bears the highest weightage of 53.4% of capital being recommended to invest in Alphabet Inc. The portfolio weightage is also sensible as all portfolio weights add up to 1, implying 100% of capital is allocated. The annual return for the portfolio is 25.45%, which is relatively high, similar to the highest ever recorded annual returns of the S&P 500 fund; however, its annualised risk is also 25.082%, implying a decently high risk of investment. The annualised Sharpe ratio of 0.9439, which implies that for every one unit of risk you take, you receive about 0.9 units of reward, which is not the most ideal Sharpe Ratio, but it is relatively adequate.

VI. DISCUSSION

A. Strengths

The ARIMA model allowed us to be able to see that the mean returns remain relatively stable around 0, which helped simplify the model and allowed for direct computation of GARCH on log returns rather than on ARIMA residuals, ensuring a faster computation without compromising heavily on results.

GARCH allows for understanding of volatility in seemingly constant mean data, allowing for a more rigorous understanding of how volatile markets can affect the log return data, which makes it extremely valuable for forecasting random shocks to the system, allowing it to better represent extreme events.

Monte Carlo Simulations and Geometric Brownian Motion simulations have allowed for a simulation of many possible paths that the stock price can take, allowing for a very large range of dynamic futures of the market, which are often the most likely path for the growth of stocks.

Modelling the unified movement with stocks allows us to also identify which markets are thriving within the portfolio as of recent, and we can make adjustments based on this movement. This allows for a high-level overview of sectors and their profitability within financial markets.

B. Weaknesses

The ARIMA model cannot be used to forecast heteroskedastic data, as it cannot perform mean calculations over data akin to white noise, meaning that, on its own, ARIMA as a model is not proficient enough to simulate data effectively.

GARCH volatility does manage to encapsulate the pattern of modelling volatility in terms of where the data is likely to be volatile; however, it does not grasp the magnitude of the volatility as well, making it slightly less effective for proper simulation of very extreme events.

We can only model extreme events to a better degree with distributions such as the Student's T distribution; however, even with this, we cannot always model black swan events (rare extreme events which have widespread consequences, such as COVID-19 or the 2008 financial crisis) that effectively and without a very large number of simulations we will not be able to get all a good representation of most outcomes, this in turn bears computational expense and black swan events are inherently unpredictable.

Using a simple covariance matrix to account for the similar movement of similar assets does not consider any differences that may arise through different objectives and areas occupied by companies; as such, assuming a constant similar movement is rather naive.

C. Potential Improvements

EGARCH (Exponential GARCH) can be used to allow GARCH models to better understand logarithmic return data, allowing for more accurate approximations of extreme volatilities.

DCC-GARCH (Dynamic Conditional Correlation GARCH) can be used to better model complex relationships between companies within the portfolio, rather than assuming a constant relationship between the company and allowing for forecasting to be based on the assumption that the companies constantly delve into the same fields and always achieve similar performance.

Regime switching models can also be implemented, rather than assuming the underlying dynamics of markets are consistently the same, which allows for market behaviours to differ with factors like season (for example, in winter, Coca-Cola company profits will not be as high as they would be in summer due to a reduction in demand based on season). This can help GARCH models capture structural breaks better.

LSTM or transformer models can also be used for mean forecasting instead of ARIMA, as they prioritise memorising sequential order heavily. [5]

VII. CONCLUSION

This project investigated the application of time series analysis and Monte Carlo methods towards forecasting stock prices and constructing optimal investment portfolios from real-world market data.

Through the usage of ARIMA(1,0,0) mean forecasting, we were able to view trends within the mean log returns across the forecasted horizon. Alongside this, we were able to identify the limitations of the ARIMA model with heteroskedastic data, thus justifying why we needed GARCH models instead of forecasting simulations through ARIMA with random normal shocks.

GARCH(1,1) allowed for volatility forecasting, which in turn allowed us to understand the stochastic nature of financial markets and, as a consequence, be able to effectively grasp the nature of markets during volatile periods (albeit in a limited manner), which allows us to predict rather peculiar black swan events.

We were able to use Monte Carlo simulations generated by random shocks to the mean log returns based on the forecasted volatilities, resulting in the development of a model that can attempt to simulate extreme events. Finally, from this, we were able to optimise the expected risk and return to obtain a risk-adjusted portfolio with a 25.4% annual rate of return with a Sharpe ratio of 0.94, allowing for an acceptable return-to-risk ratio with high annual return.

Despite the successes, there remain several limitations on our forecasting due to the unpredictable nature of financial markets that depend on a plethora of factors. As such, assumptions such as constant mean and consistent time-varying volatility do not hold in cases such as changes in regimes, announcement of macroeconomic policies. To even attempt to account towards such factors, real-time information is required as well as machine learning models that can account for complex relationships of volatility or between assets; additionally, backtesting methods can also be used to determine the effectiveness of the portfolio.

VIII. ANNEX

A. Numerical Integration Methods

Numerical integration methods are used heavily in orbital astrophysics and spatial engineering, as calculating gravitational interactions between multiple bodies, as well as calculating orbital dynamics, relies on different integration methods (such as Velocity Verlet integration). As the number of interacting objects increases, analytical solutions tend to become impossible to solve on the scale of astrophysics with a force as extreme as gravity (extremely weak at longer distances and extremely strong at shorter distances). Oftentimes, Runge-Kutta integration methods are used as well for approximating orbits. [6]

Numerical Integration methods are used to address material composition and strength, as well as to stress test the structural integrity of materials in structural engineering. Monte Carlo Integral solutions are used to demonstrate how fissures can

occur and expand upon impact, provided frontal crashes occur in the material. [7]

Complex quantum mechanical systems often use numerical methods to solve for probability density functions where analytical solutions are not possible, such as a quantum anharmonic oscillator simulation. To simulate the time evolution of the Schrodinger Equation, you require numerical Runge-Kutta 4th or 5th order integrations or Geometric Numerical Integration (to conserve fundamental quantities), which can allow you to treat the equation as a system of simpler differential equations that can be solved for the time evolution of the Schrodinger Equation in complex quantum mechanical systems. [8]

The aerospace industry utilises numerical integration methods such as leapfrog integration or velocity verlet integration to simulate aircraft orbital path dynamics, while geometric integration methods and finite volume difference methods are used for aerodynamics of aircraft. Flight simulators have to account for all of the above factors when considering spatial missions and even aircraft departures. [9]

Climate modelling requires numerical integration methods to approximate differential equations in thermodynamics of the climate (free energy and entropy estimations) and fluid dynamics (finite difference integration methods). These approximations are used for long-term climate predictions and weather forecasting when combined with time series analysis. [10]

Banks and Hedge funds utilise Monte Carlo integration methods and quadrature methods for the pricing of complex options, alongside credit and interest rate derivatives. These methods are essential for asset management alongside modelling financial markets and considering complex market dynamics. [11]

B. Runge Kutta Methods

Non-periodic flow systems rely on Runge-Kutta Methods to approximate solutions. Such systems are heavily sensitive and dependent on initial value conditions, which is why Runge-Kutta approximation methods are used to model such nonlinear dynamical systems while also attempting to simulate the long-term evolution of the system. [12]

Electrical engineering software using Runge-Kutta 4th and 5th order simulations to model the time evolution of both linear and nonlinear circuit systems. Often utilised in software such as MATLAB Simulink, which uses RK45 solvers to test the effectiveness of complex circuits over long periods of time. [13]

Modelling ecosystems often requires solving coupled differential equations with initial values to see how predators and prey interact with one another, and breeding interactions (how populations are controlled within ecosystems. Runge-Kutta 4th-order approximation methods are used due to their remarkable accuracy without requiring computational power, as RK4 methods have $O(N)$ time complexity, making them ideal for ecological system simulation. Most notably used in the Lotka-Volterra equations that describe prey-predator interactions. [14]

Simulation of movements and navigation systems depends heavily on differential equations, which require low-latency and limited-memory code for high-accuracy approximations, which is where RK4 solvers with $O(N)$ time complexity are used for simulating movement from inverse kinematic equations. These RK4 solvers can be run natively on microcontrollers with some floating-point approximations, making RK4 solvers ideal for solving dynamic movement equations in autonomous motion. [15]

Nuclear and plasma physics require solving equations for particle motion in directed electromagnetic fields, which is necessary for modelling confinement systems in fusion reactors to ensure the safety of the reactor. Electron plasma beams can also be confined within these fields; this can be modelled with RK4 solvers to model the density distributions. [16]

Launch trajectory simulations often require RK4 solvers to approximate differential equations that need to account for thrust, gravitational forces, and fuel consumption (especially since fuel consumption is dynamic and affects both the mass and thrust of the aircraft). These simulations are as they are essential for launch trajectory planning. [17]

C. Fourier Transforms and Time Series Analysis

Fourier series are essential in nuclear physics. Fourier transforms of charge distributions can be utilised to solve for the charge density of atomic nuclei and can be used to adjust for effects of pulses on fermions (spin 1/2 particles) and their interactions within strong electromagnetic fields (of radio frequency, RF). [18]

Time series analysis techniques and Fourier transforms are used to process signal data from seismic wave frequencies and patterns caused by earthquakes. Periodic structures and behaviour are analysed to help predict geological behaviour and identify fault lines as well. [19]

MRI scanners use Fourier transforms to reconstruct spatial images or the frequency domain of radio wave measurements. These transforms are used to reduce noise and increase image clarity for diagnostics. [20]

Communication systems use fast fourier transforms for signal compression and bandwidth optimisation techniques and frequency domain analysis is often essential for noise reduction in 4G and 5G data transfers. [21]

Factories utilise time series methods and Fourier analysis to account for frequency patterns in breakdowns and prevent breakdowns before they happen, or schedule maintenance earlier depending on previously reported data. [22]

CODE AVAILABILITY

The full implementation for this project is The full implementation used in this project is available in the following Public GitHub repository:

<https://github.com/Krishk1223/Stock-volatility-and-trend-analysis>

The repository contains:

- Python scripts for initial configuration, data processing, time series analysis, model development and simulation.
- Jupyter Notebooks that interactively use utilities from the scripts to help justify the reasoning for the choices made in this paper.
- Preliminary instructions for this repository's usage.

REFERENCES

- [1] E. J. Elton, M. J. Gruber, S. J. Brown, and W. N. Goetzmann, *Modern Portfolio Theory and Investment Analysis* (John Wiley & Sons, 2009).
- [2] A. A. Ariyo, A. O. Adewumi, and C. K. Ayo, *Stock Price Prediction Using the ARIMA Model*, in *2014 Uksim-AMSS 16th International Conference on Computer Modelling and Simulation* (2014), pp. 106–112, ?.
- [3] T. Bollerslev, Generalized autoregressive conditional heteroskedasticity, *Journal of Econometrics* **31**, 307 (1986).
- [4] P. Jorion, Portfolio optimization in practice, *Financial Analysts Journal* **48**, 68 (1992).
- [5] Z. Zhang, B. Chen, S. Zhu, and N. Langrené, Quantformer: from attention to profit with a quantitative transformer trading strategy, *Expert Systems with Applications* **313**, 131567 (2026).
- [6] O. Montenbruck, Numerical integration methods for orbital motion, *Celestial Mechanics and Dynamical Astronomy* **53**, 59 (1992).
- [7] S. Boria, J. Obradovic, and G. Belingardi, Experimental and numerical investigations of the impact behaviour of composite frontal crash structures, *Composites Part B: Engineering* **79**, 20 (2015).
- [8] E. Faou, *Geometric Numerical Integration and Schrödinger Equations*, Vol. 15 (European Mathematical Society, 2012).
- [9] R. Howe, *An Improved Numerical Integration Method for Flight Simulation*, in *Flight Simulation Technologies Conference and Exhibit* (1989), p. 3306, ?.
- [10] G. Mengaldo, A. Wyszogrodzki, M. Diamantakis, S.-J. Lock, F. X. Giraldo, and N. P. Wedi, Current and emerging time-integration strategies in global numerical weather and climate prediction, *Archives of Computational Methods in Engineering* **26**, 663 (2019).
- [11] P. Capuozzo, E. Panella, T. S. Gherardini, and D. D. Vvedensky, Path integral Monte Carlo method for option pricing, *Physica A: Statistical Mechanics and Its Applications* **581**, 126231 (2021).
- [12] E. N. Lorenz, Deterministic nonperiodic flow 1, *Universality in Chaos, 2nd Edition* 367 (2017).
- [13] P. Maffezzoni, L. Codecasa, and D. D'Amore, Time-domain simulation of nonlinear circuits through implicit Runge–Kutta methods, *IEEE Transactions on Circuits and Systems I: Regular Papers* **54**, 391 (2007).
- [14] S. Paul, S. P. Mondal, and P. Bhattacharya, Numerical solution of Lotka Volterra prey predator model by using Runge–Kutta–Fehlberg method and Laplace Adomian decomposition method, *Alexandria Engineering Journal* **55**, 613 (2016).
- [15] M. D. Fiore and C. Natale, *Discrete-Time Closed-Loop Inverse Kinematics: A Comparison between Euler and Rk4 Methods*, in *2021 29th Mediterranean Conference on Control and Automation (MED)* (2021), pp. 584–589, ?.
- [16] T. Matsui, D. Adachi, S. Iwata, T. Urano, and T. Takahashi, Full-particle simulation on beam electron plasma in linear confinement system, *Plasma and Fusion Research* **13**, 3401041 (2018).
- [17] R. Mollo, Modeling and Trajectories Simulation for Complex Aerospace Systems, Doctoral dissertation, 2019, ?.
- [18] N. Krishna and S. Gordon, Fourier Transform for a Single Nucleus of Spin $\frac{1}{2}$ in the Presence of a Strong rf Field, *Physical Review A* **6**, 2059 (1972).
- [19] E. A. Robinson, Spectral approach to geophysical inversion by Lorentz, Fourier, and Radon transforms, *Proceedings of the IEEE* **70**, 1039 (1982).
- [20] T. A. Gallagher, A. J. Nemeth, and L. Haccin-Bey, An introduction to the Fourier transform: relationship to MRI, *American Journal of Roentgenology* **190**, 1396 (2008).
- [21] J. Salz and S. B. Weinstein, *Fourier Transform Communication System*, in *Proceedings of the First ACM Symposium on Problems in the Optimization of Data Communications Systems* (1969), pp. 99–128, ?.
- [22] S. Patil and J. Gaikwad, *Vibration Analysis of Electrical Rotating Machines Using FFT: A Method of Predictive Maintenance*, in *2013 Fourth International Conference on Computing, Communications and Networking Technologies (ICCCNT)* (2013), pp. 1–6, ?.